Impact Of Gender and Multimodal Integration on Biometric Authentication Systems

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Abstract

This report investigates the influence of gender on the performance of voice and face recognition systems and evaluates the efficacy of a multimodal authentication system that integrates these two biometrics. We hypothesize that gender differences exist in the accuracy of biometric authentication and that a multimodal approach can enhance overall system robustness and accuracy. Preliminary results suggest that gender impacts recognition performance, with female data exhibiting higher accuracy, and that the multimodal system outperforms the unimodal systems, indicating a beneficial integration effect.

1 Introduction

Biometric authentication systems utilize unique physiological characteristics to identify individuals. In the realm of security, such systems have become increasingly critical due to their non-repudiation and difficulty to forge. This report delves into the construction and evaluation of two singular modality biometric systems—one based on voice recognition and the other on facial landmarks—and a multimodal system that integrates both modalities.

The research questions guiding this study are:

1. How does gender impact the performance of voice and face recognition systems?
2. How does the integration of voice and facial recognition systems enhance authentication accuracy compared to using these modalities independently?

The biometric data utilized in this study were obtained from publicly available sources and processed using a combination of RandomForestClassifier and K-Fold Cross-Validation techniques. The decision-level fusion was employed in the multimodal system to capitalize on the strengths of each modality. The initial analysis indicates a higher accuracy in recognition for female subjects compared to male subjects.

2 Datasets

The construction of the biometric authentication systems required the collection and preparation of datasets for both voice and facial recognition. This section provides a detailed description of the datasets used, the sources from which they were obtained, and their respective preprocessing steps.

2.1 Voice Dataset

The voice dataset was acquired from the Sample Voice Data repository hosted on GitHub, which provides a collection of voice recordings split by gender. This dataset facilitates the study of gender differences in voice recognition systems.

- Source: Sample Voice Data available at https://github.com/jim-schwoebel/sample_voice_data
- Composition: The dataset includes a collection of male and female voice samples, organized into separate folders by gender.
- Preprocessing: Audio samples were preprocessed to normalize volume levels and to extract salient features crucial for effective voice pattern recognition.

Dataset Feature	Details
Source	Sample Voice Data from GitHub
URL	https://github.com/jim-schwoebel/sample_voice_data
Content	Audio recordings of male and female voices
Organization	Data split into separate folders by gender
Data Type	Audio Samples
Usage	Voice Recognition
Data Collection	Zip file downloaded from the repository

Table 1: Summary of the voice dataset used in the biometric authentication system project.

Both were divided into training and testing sets to facilitate the development and evaluation of the biometric systems.

2.2 Face Dataset

The face dataset was selected from a Kaggle repository containing segregated images based on gender. 1,500 photos in .png format were selected from each folder and used for our research.

3 features

Our research is separated into three modalities: face recognition, voice recognition, a multimodal recognition system incorporating both face and voice. The first step was to determine the accuracy of voice recognition alone when comparing the two genders. A flexible programming language would be needed to achieve the results needed for the tasks. Python was the language chosen for all the coding and analysis.

3.1 Voice Comparison

To start, features were extracted from each audio file using Librosa.Librosa is a python package for music and audio analysis. Once features were successfully extracted, both male and female data were split into training and test sets using train split test imported from sklearn.model_selection module. The is a python package from scikit-learn that provides simple and predictive tools for data analysis. The overall accuracy of the model was determined

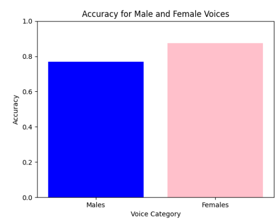

Figure 1: Voice Analysis

first. The classifier "RandomForestClassifier" was trained and implemented into our process. This was used to predict labels for the test set. The accuracy for recognition of male voices was 0.769 while the females scored an accuracy rate of 0.875. This discovery became an expectation for our upcoming research, but to our surprise, it varied.

Next the equal error rate(EER) was determined. Equal Error Rate (EER) is a metric used in biometric security systems to measure the effectiveness of the system in identifying individuals correctly. EER is the point at which the false acceptance rate (FAR) and false rejection rate (FRR) are equal [?]. The csv file was a key component in delivering the information needed. By processing the csv file, features and its respective labels were taken and split into training and test sets. Random Forest classifier was initialized and produced the model accuracy, along with the confusion matrix. To caculate the EER, the false positive (FPR), false negative rates (FNR), and threshold needed to be initialized. we had a model accuracy of 0.64 and an EER: 0.37 at threshold 0.54.

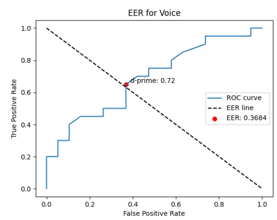

Figure 2: Voice Recognition

3.2 Face Comparison

To manipulate this model, we extracted features from images located in a specified directory.The program read images from a list of files and extracted features using the Histogram of Oriented Gradients (HOG) method. After this process, the script proceeds to compute distances between features and use these distances as inputs for machine learning models. A Nearest-Centroid classifier is employed for classification, trained using k-fold cross-validation to evaluate gender classification performance. Separate accuracy metrics are calculated for male and female images to assess any differences in classifier performance by gender.A function was also created to calculate pairwise Euclidean distances between feature vectors. Then kfold cross validation was setup to evaluate the model. After combining and shuffling the data, the script ensures that training and testing data are stratified by gender to maintain a balanced representation. The accuracy score for the male samples was 0.710 while the female samples arrived at a score of 0.65 The EER was calculated, along with D-prime show below. This amounts to an EER of 3.21 at threshold .47 giving an .82% accuracy. This outcome surpassed expectation, and brought forth an even bigger challenge for our multi-modal system.

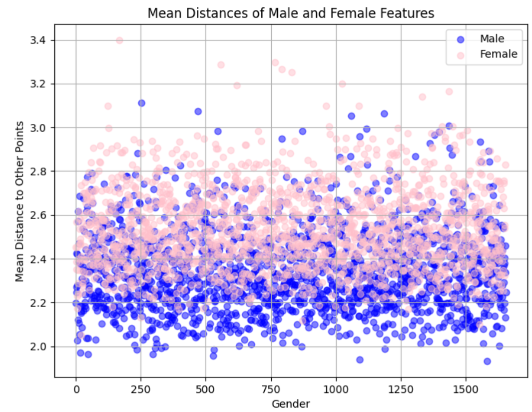

Figure 3: Mean Distances For facial Features

4 Methods

The last step was to integrate face and voice recognition into a dual system that conceptualizes data processing and feature extraction. This is create a authentic gender classification algorithm using audio and visual cues. Librosa was again used to gather Mel-frequency cepstral coefficients (MFCCs), a common set of features used in audio processing, from .wav files. The mean vectors taken from MFCCs are stored as features.As for face feature extraction, each image is again read in gray-scale, resized to standard dimensions, and processed to extract Histogram of Oriented Gradients (HOG) features. To ensure uniformity among feature vectors, which is critical for many machine learning models, a function was created to standardize the length of each feature vector. Then both male and female features and labels from voice and face data are conjoined. Labels were assigned(0 = male, 1 = female) and features are concatenated to form a complete dataset ready for training a machine learning model. The machine learning

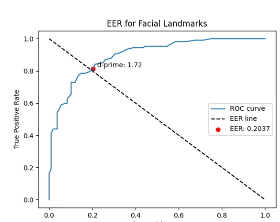

Figure 4: EER for Face Landmarks

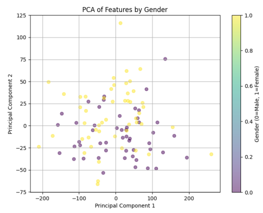

Figure 5: PCA of Features by Gender

process used is RandomForestClassifier for gender classification. The process involves training the model on a dataset where features and labels are derived from male and female data, and then evaluating the model's accuracy on these distinct groups separately, as well as on a combined test set. Data for males and females is split separately into training and testing subsets using the train_test_split function.

4.1 Training Statistics

- test_size is set to .2 reversing this amount for testing.
- random_state set to 42, guaranteeing that the splits are identical on each run.
- estimators is 50 meaning the model will use 50 trees in the forest.
- min_samples_leaf is 5. Each leaf must have at least 5 samples.

The male accuracy was 70.00% female accuracy was50.00% with an overall accuracy: 60.00%. To our surprise, it scored lower than the others.After calculations, the EER was .238.

5 Conclusion

While the concept of multi-modal biometric systems, which combine data from multiple sources such as face and voice recognition, is often touted for its potential to enhance accuracy and security, these systems are not universally superior to their uni-modal counterparts. Integration enhances accuracy over voice recognition alone but doesn't exceed facial recognition.

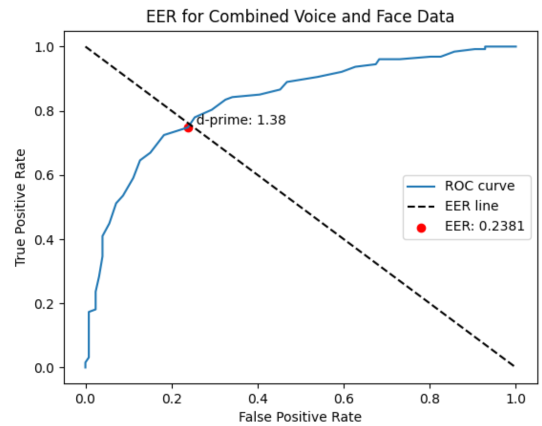

Figure 6: EER for Voice and Face Data

6 References

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